What’s Wrong With Science?

A University Course

Professor David Gresham
Fall 2026

University Courses

With so many courses to choose from each semester, it can be hard to find space for something new, especially something that offers a chance to step outside your major and explore pressing issues, from freedom of expression to the future of food. That's where NYU's new 1-credit University Courses come in.

University Courses are large interdisciplinary classes that invite undergraduate students from across NYU to engage with pressing societal issues (e.g., climate change, democracy, artificial intelligence) and current events. **Open to undergraduates from any school or discipline**, these short, pass/fail courses are designed for students who are eager to connect across fields and explore timely topics. Most courses meet just 1.5 to 2 hours per week, making them an easy and engaging way to broaden your perspective. **No prerequisites, just curiosity.**

Professor David Gresham - an actual real life scientist

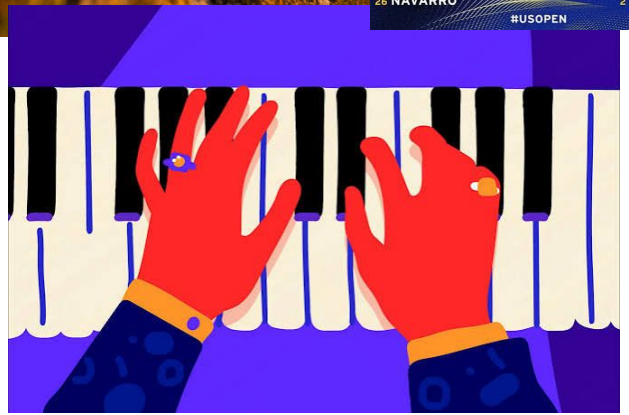






Professor David Gresham - also a human being











BSc: Major in Biochemistry, Minor in Archaeology



PhD in Human Genetics, Edith Cowan University







Am. J. Hum. Genet. 67:47–58, 2000

N-myc Downstream-Regulated Gene 1 Is Mutated in Hereditary Motor and Sensory Neuropathy–Lom

Luba Kalaydjieva,[1,2] David Gresham,[1,*] Rebecca Gooding,[1,*] Lisa Heather,[1] Frank Baas,[3] Rosalein de Jonge,[3] Karin Blechschmidt,[4] Dora Angelicheva,[1] David Chandler,[1] Penelope Worsley,[1] Andre Rosenthal,[4] Rosalind H. M. King,[5] and P. K. Thomas[5]

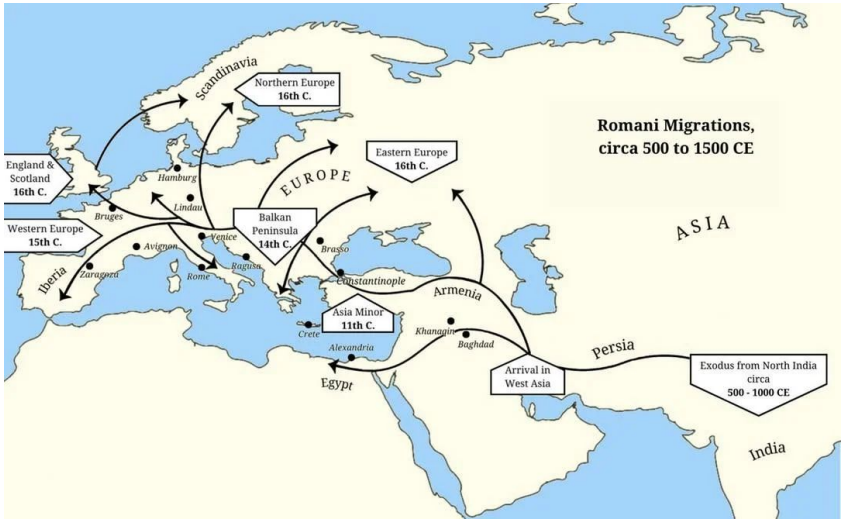
[1]Centre for Human Genetics, Edith Cowan University, and [2]Western Australian Institute for Medical Research, Perth, Australia; [3]Neurozintuigen Laboratory, Academic Medical Centre, University of Amsterdam, Amsterdam; [4]Institute of Molecular Biotechnology, Jena, Germany; and [5]Institute of Neurology, University College London, London

Am. J. Hum. Genet. 69:1314–1331, 2001

Origins and Divergence of the Roma (Gypsies)

David Gresham,[1] Bharti Morar,[1] Peter A. Underhill,[3] Giuseppe Passarino,[4] Alice A. Lin,[3] Cheryl Wise,[1] Dora Angelicheva,[1] Francesc Calafell,[5] Peter J. Oefner,[6] Peidong Shen,[6] Ivailo Tournev,[7] Rosario de Pablo,[9] Vaidutis Kučinskas,[10] Anna Perez-Lezaun,[5] Elena Marushiakova,[8] Vesselin Popov,[8] and Luba Kalaydjieva[1,2]

[1]Centre for Human Genetics, Edith Cowan University, and [2]Western Australian Institute for Medical Research, Perth; [3]Department of Genetics, Stanford University School of Medicine, Stanford, CA; [4]Dipartimento di Biologia Cellulare, Università della Calabria, Rende, Italy; [5]Unitat de Biologia Evolutiva, Facultat de Ciencies i de la Vida, Universitat Pompeu Fabra, Barcelona; [6]Stanford Genome Technology Center, Palo Alto, CA; [7]Department of Neurology, Medical University, and Foundation for Health Problems of Ethnic Minorities, and [8]Institute of Ethnology, Bulgarian Academy of Sciences, Sofia; [9]Unidad de Immunologia, Clinica Puerta de Hierro, Madrid; and [10]Human Genetics Centre, Medical Faculty, University of Vilnius, Vilnius, Lithuania




nature
genetics
volume 33
march 2003
A ten-year retrospective
1992–2002

Postdoctoral Fellowship



PRINCETON UNIVERSITY

Genome-Wide Detection of Polymorphisms at Nucleotide Resolution with a Single DNA Microarray

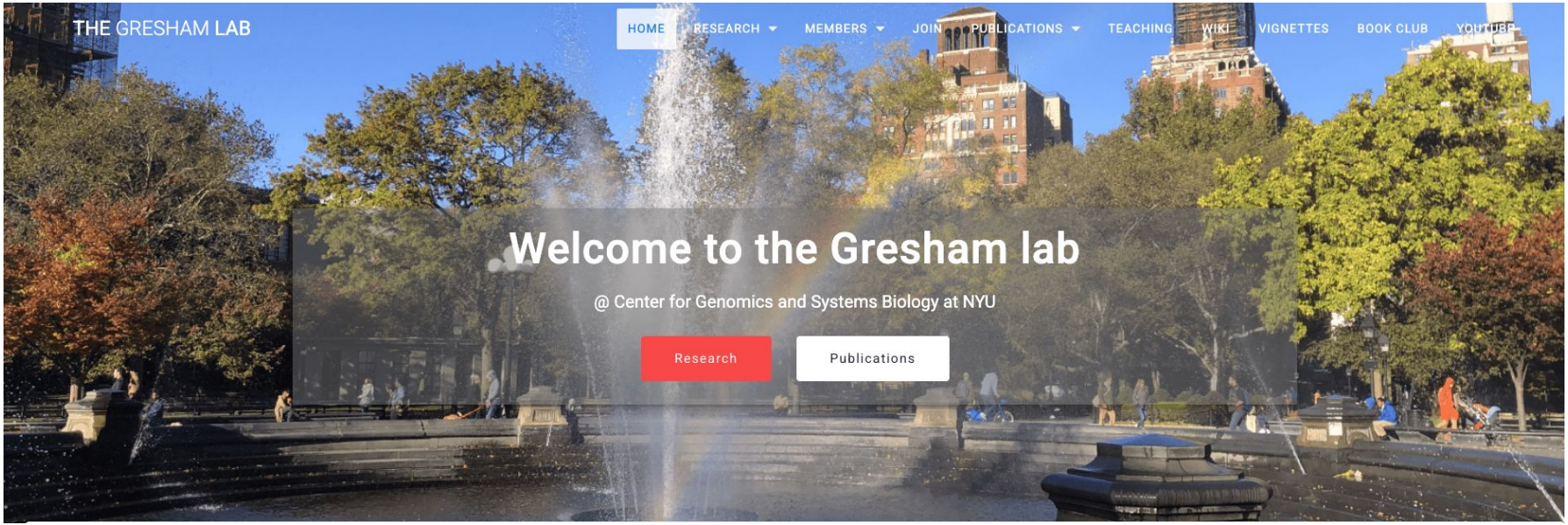
David Gresham,[1,2*] Douglas M. Ruderfer,[1,3] Stephen C. Pratt,[1,3] Joseph Schacherer,[1,3] Maitreya J. Dunham,[1] David Botstein,[1,2] Leonid Kruglyak[1,3*]

OPEN ACCESS Freely available online PLoS GENETICS

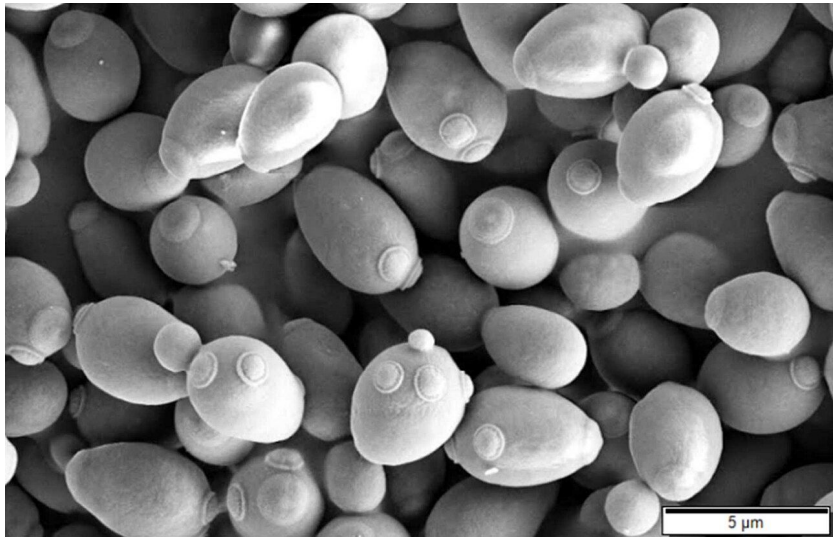
The Repertoire and Dynamics of Evolutionary Adaptations to Controlled Nutrient-Limited Environments in Yeast

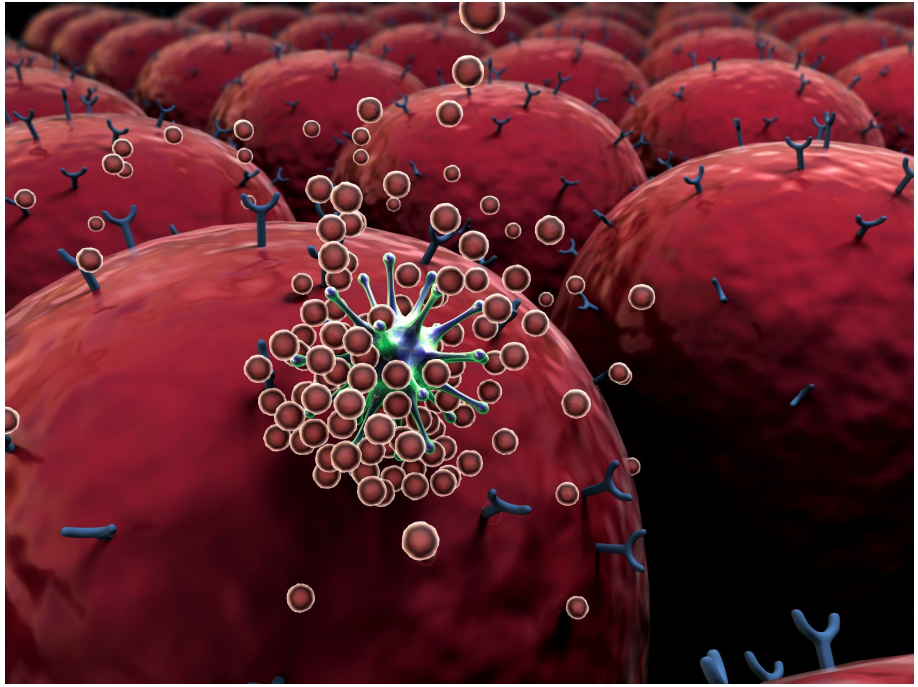
David Gresham[1,2], Michael M. Desai[1], Cheryl M. Tucker[1], Harry T. Jenq[1], Dave A. Pai[1], Alexandra Ward[1], Christopher G. DeSevo[1], David Botstein[1,2*], Maitreya J. Dunham[1,3*]

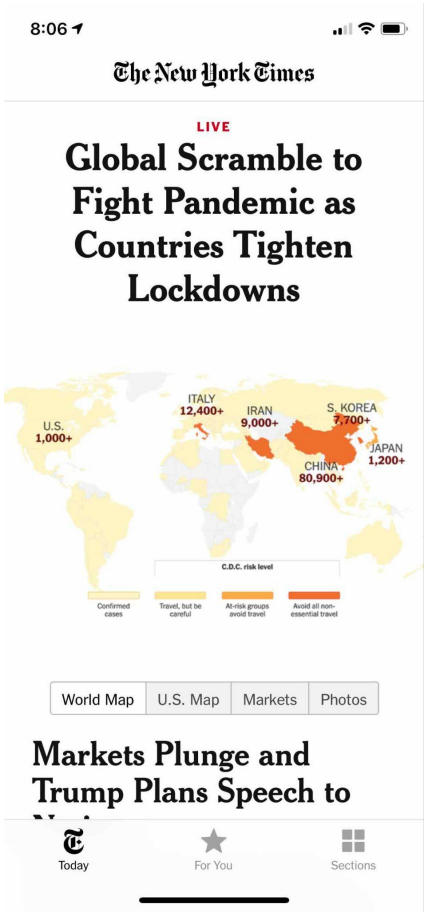
The Gresham Lab at NYU (founded 2009)



We research genome evolution in microbes






8:06
The New York Times
LIVE
Global Scramble to Fight Pandemic as Countries Tighten Lockdowns
U.S. 1,000+
ITALY 12,400+
IRAN 9,000+
S. KOREA 7,700+
JAPAN 1,200+
CHINA 80,900+
C.D.C. risk level
Confirmed cases
Travel, but be careful
At-risk groups avoid travel
Avoid all non-essential travel
World Map
U.S. Map
Markets
Photos
Markets Plunge and Trump Plans Speech to
Today
For You
Sections


THIS STORE IS TEMPORARILY
AUTOMATIC CAUTION DOOR
ATTENTION CUSTOMERS
All firearms have been removed from the sales floor.
Thank you.
$40 BONUS REWARD
Store Open for Curbside Pick-up Only
9am-6pm
FOR CURBSIDE PICKUP
845-382-2151
NOW AVAILABLE
CURBSIDE PICKUP
PLEASE CALL
845-382-2151
DICKS.COM/MORE
AUTOMATIC CAUTION DOOR



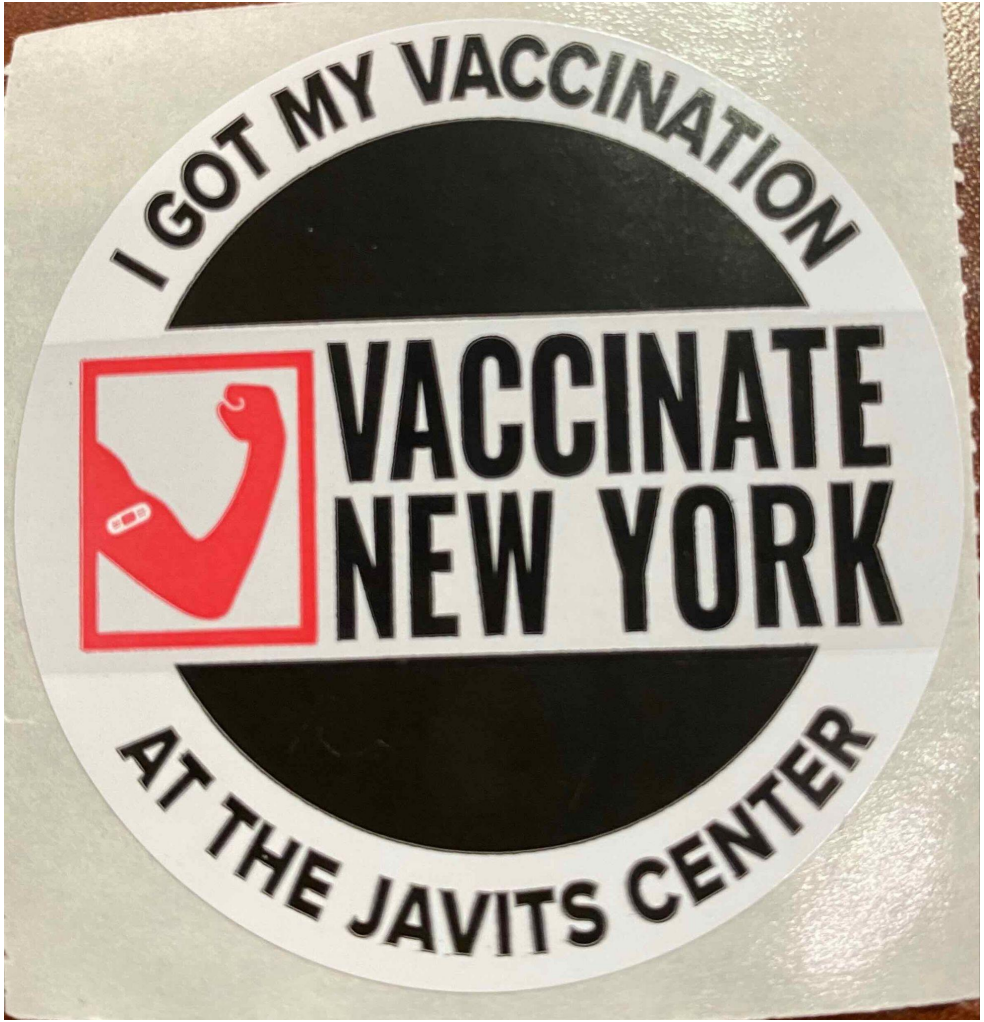
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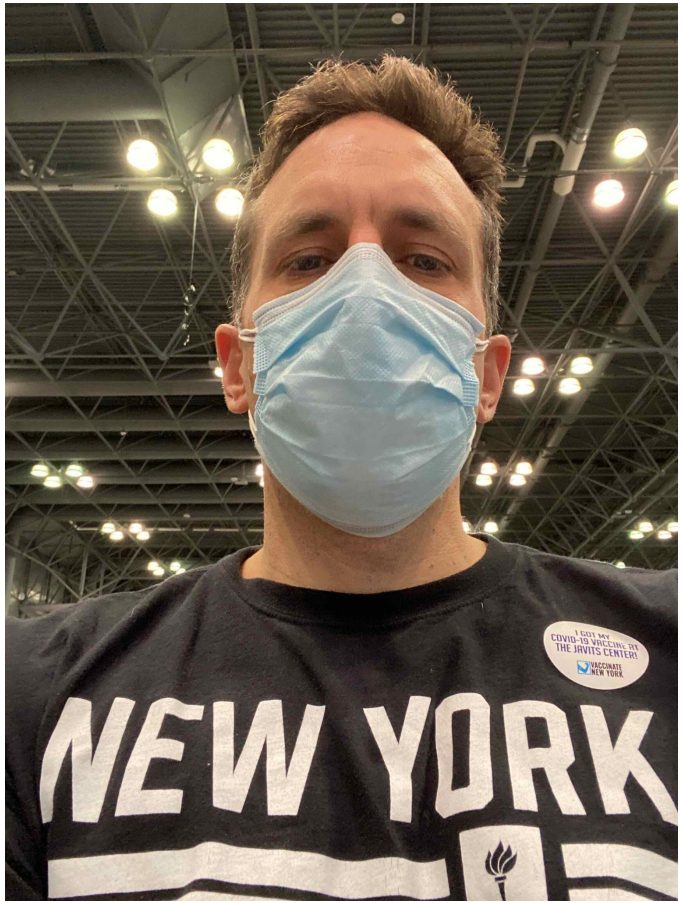
Back

NYU

YOU MAY REPORT TO WORK




I GOT MY VACCINATION
VACCINATE
NEW YORK
AT THE JAVITS CENTER


I GOT MY COVID-19 VACCINE AT THE JAVITS CENTER!
VACCINATE NEW YORK
NEW YORK

March 8, 2021

THE NOBEL PRIZE IN PHYSIOLOGY OR MEDICINE 2023



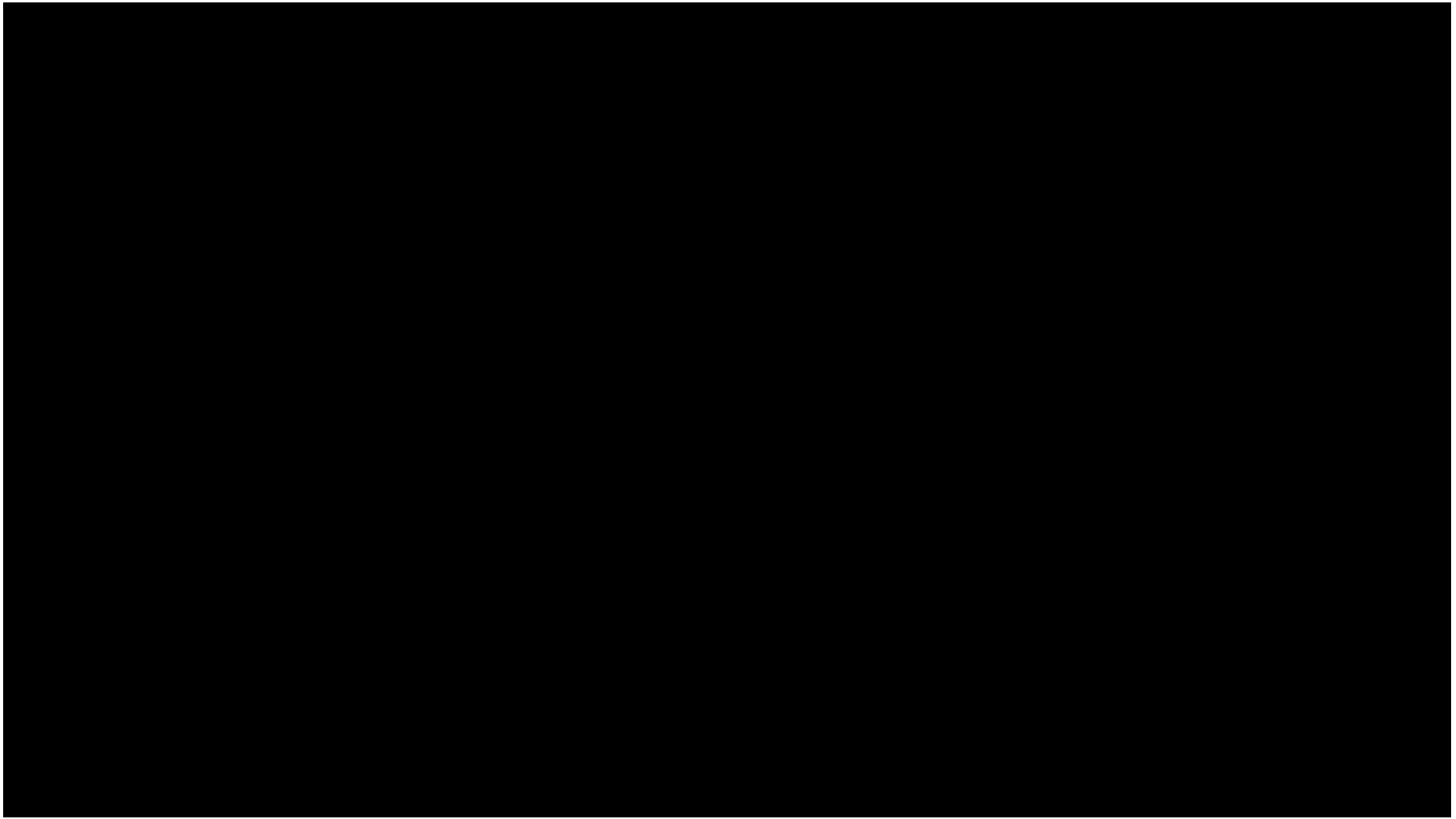
Illustrations: Niklas Elmehed

Katalin Karikó Drew Weissman

"for their discoveries concerning nucleoside base modifications that enabled the development of effective mRNA vaccines against COVID-19"

THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET


OPERATION
WARP SPEED
OPERATION
WARP SPEED
gettyimages
Credit: The Washington Post
2229385358



CHILDREN'S HEALTH

Trump and RFK Jr. try again to upend vaccines for kids

UPDATED AUGUST 10, 2026 · 6:42 PM ET

HEARD ON ALL THINGS CONSIDERED

Rob Stein

3-Minute Listen + PLAYLIST TRANSCRIPT ···



Idaho Capital Sun

HEALTH

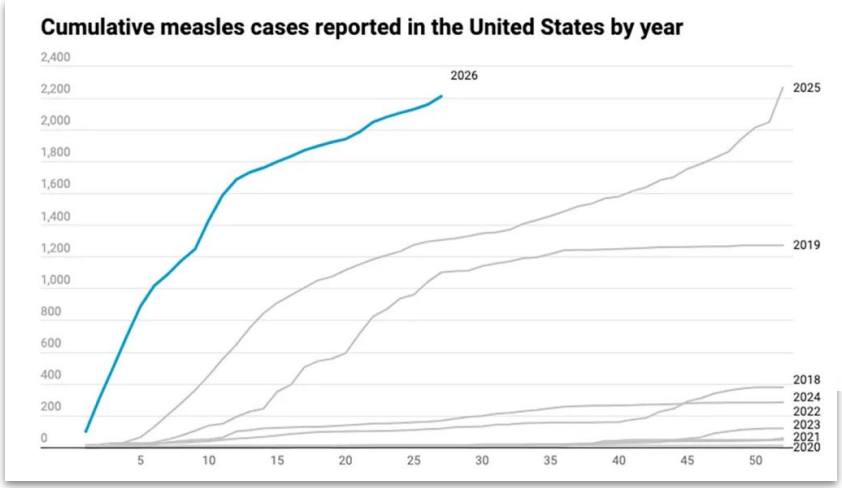
Health Department declares measles outbreak in Wyoming's Teton County amid influx of tourists

Wyoming reports third case amid heavy influx of tourist crowds in Jackson and Grand Teton National Park

BY **KATIE KLINGSPORN** - JULY 13, 2026 4:54 PM

The Wyoming Department of Health on Monday announced an outbreak of measles in Teton County amid a heavy influx of summer visitors.

The department declared the outbreak after confirming a third case of the highly infectious illness in Teton County this summer. The public may have been exposed at several locations in Jackson and neighboring Grand Teton National Park.



Measles Cases Hit New Record in U.S., as Vaccination Rates Wane

More measles cases have been reported in the United States in the last two years than in all the years from 2000 through 2024 combined.

Listen · 6:06 min Share full article 854



A patient with measles symptoms in South Carolina in January. Getty Images

By Teddy Rosenbluth

July 24, 2026

THE WALL STREET JOURNAL.

HEALTH • WELLNESS

Teen Boys and Young Men Are Injecting Peptides in Search of Perfection

Injections that promise glowing skin and muscle growth have captivated ever-younger groups, including young men and even teenagers

Aa 376 Gift unlocked article Listen (9 min)

By Sara Ashley O'Brien | Photography and video by Sam Hayes for WSJ. Magazine
May 11, 2026 12:09 pm ET

For as long as he can remember, Trevor Larcom wanted to look different.

He'd grown up overweight in the public eye as a child actor, appearing on the sitcoms "Fresh Off the Boat" and "Fuller House." He was 18 years old and 300 pounds when he turned his focus to working out and cutting calories.

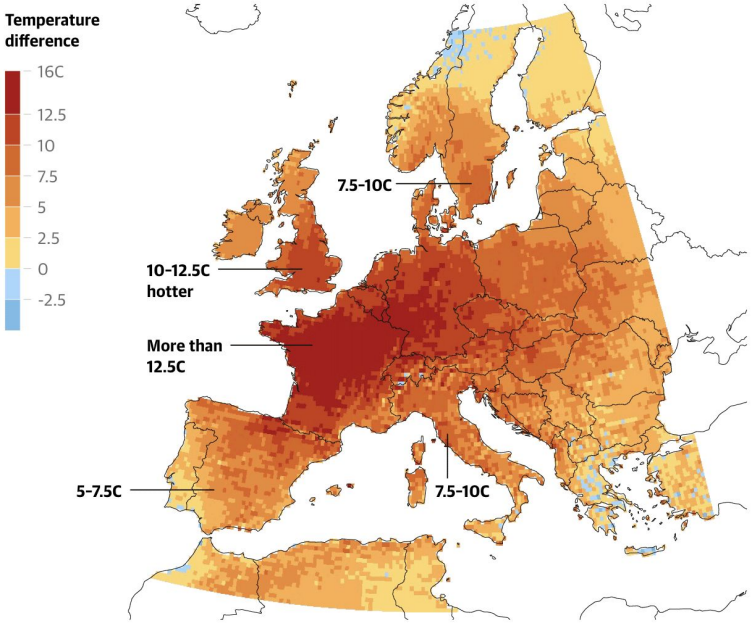
Yet once he was down to 170 pounds, he still felt insecure. Dating wasn't going well. *What if I could change how my face is presented to people?* he recalled thinking.

"One of the first things I looked up was 'how to get a better jawline,'" said Larcom, now 22.

That's how he fell into the online world of looksmaxxing, where young men relentlessly pursue physical ideals. He dyed his eyebrows. He did neck exercises and chewed extra-firm gum that he'd seen looksmaxxers claim would help build the jawline's masseter muscles. And after seeing numerous before-and-after transformations, he ordered a peptide "stack," or a combination of several peptides for supposed enhanced results.



Average daily max temperature, 22-28 Jun 2026 relative to a 1991-2020 baseline



Guardian graphic. Source: ERA5, World Weather Attribution

The heatwave has broken temperature records across Europe



Asia Australia Middle East Africa Inequality Global development

At least 35,000 excess deaths recorded in Europe's back-to-back heatwaves

True figure likely to be much higher as it only covers about half of continent and does not include August totals



NEWS | 20 August 2026

Exclusive: NSF set to issue lowest number of new grants in four decades

The US science-funding agency is withholding $1 billion of its budget so that the money can go to a special White House project.

By Dan Garisto

Scientists warn US will lose a generation of talent because of Trump cuts



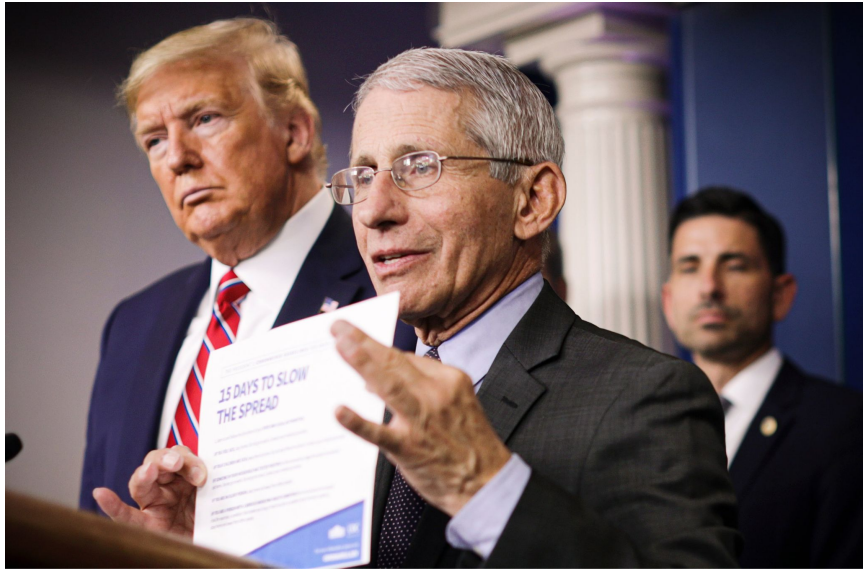
Scientists Scramble for Private Cash in Response to Fresh Funding Turbulence

Additional changes are looming, prompting more researchers to downsize labs or pursue money from outside government



Erin Ingvalson of the University of Washington put a postdoctoral researcher hire on hold when funding for the position became uncertain.

By Chao Deng |Photography by Eirik Johnson for WSJ
Aug. 18, 2026 6:00 am ET



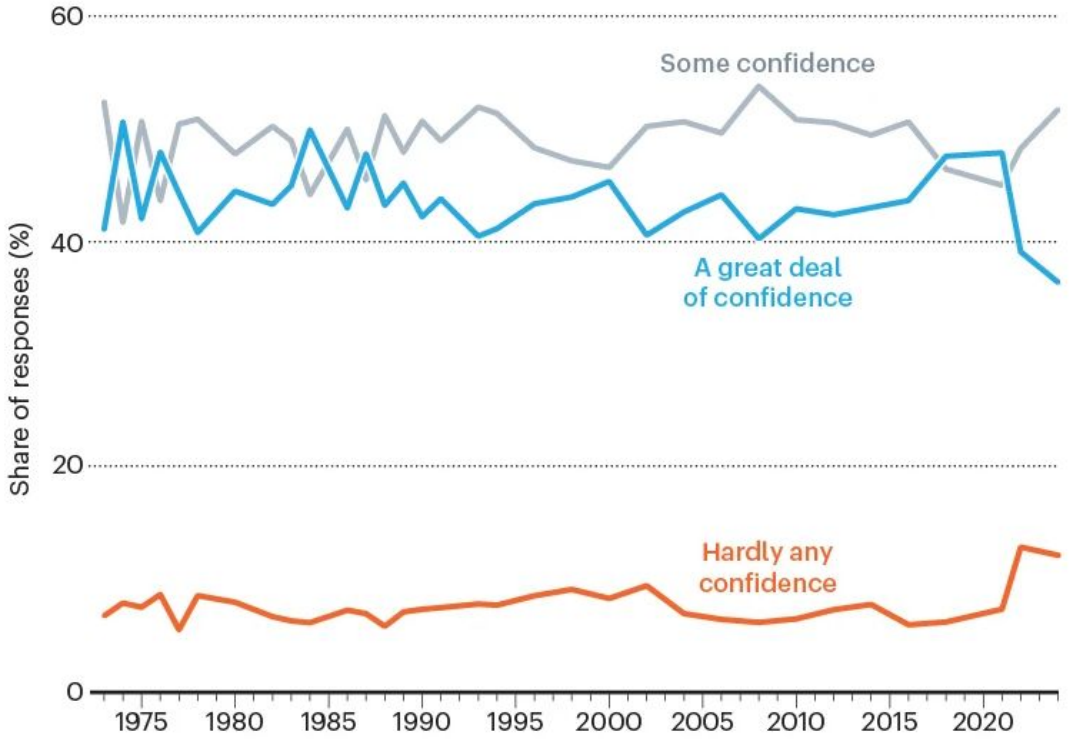
March 2020



August 2026

US CONFIDENCE IN SCIENCE

A national survey shows that people in the United States have had generally stable confidence in the scientific community over the past half-century, although the proportion with a great deal of confidence has declined sharply in the past few years.



Data from the General Social Survey. 2021 survey methods differed from other years due to the COVID-19 pandemic. ©nature

Source: General Social Survey.

BOSTON REVIEW



Foreground: White House science advisor Michael Kratsios (far left), President Donald Trump, and Office of Management and Budget director Russell Vought (far right) in June 2026. Background: President Lyndon

FORUM

How to Save Science

Trump has killed the postwar social contract, but its demise was long in the making. To protect science as a public good, we need a different approach.

Zeynep Pamuk

With responses from →

Jeremy Berg, **Alondra Nelson** *&* **Jenny Reardon**, **Madeleine Baker**, **C. Brandon Ogbunu**, **Natalie B. Aviles**, **Samuel R. Bagenstos**, **Heather Douglas**, **Robert Cook-Deegan** *&* **Arthur Daemmrich**, and **James Owen Weatherall**

Does science need to take a hard look at itself?

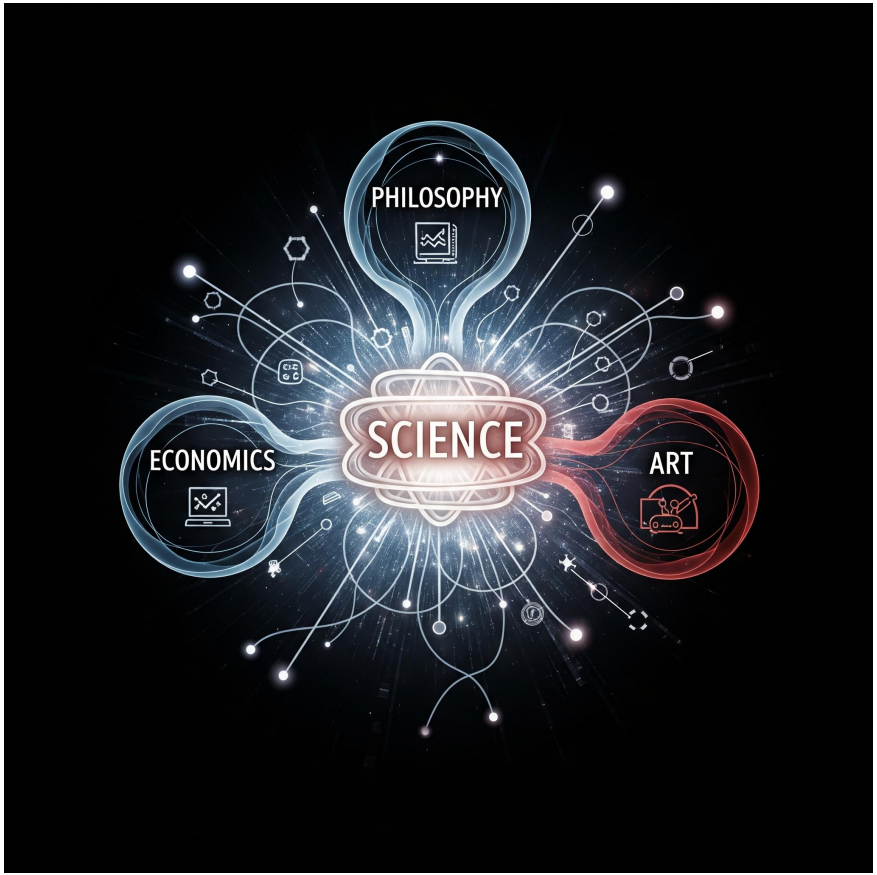


What's Wrong with Science?

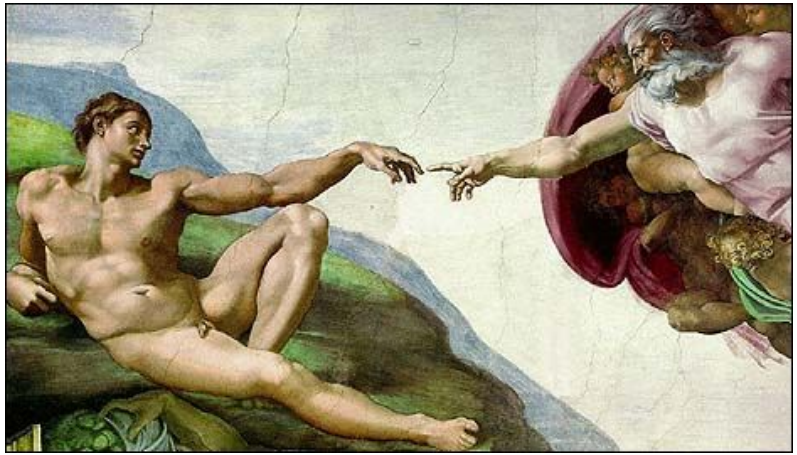
Course goals

- Explore the foundational concepts of scientific inquiry
- Examine how science is done in practice and how it progresses
- Analyze the origins of mistrust in science and its increasing polarization
- Confront the legacy of misuse of science in society
- Discuss how science impacts our lives and society
- Provide a foundation from which to contribute to enhancing the benefits of science for society

This course is for everyone

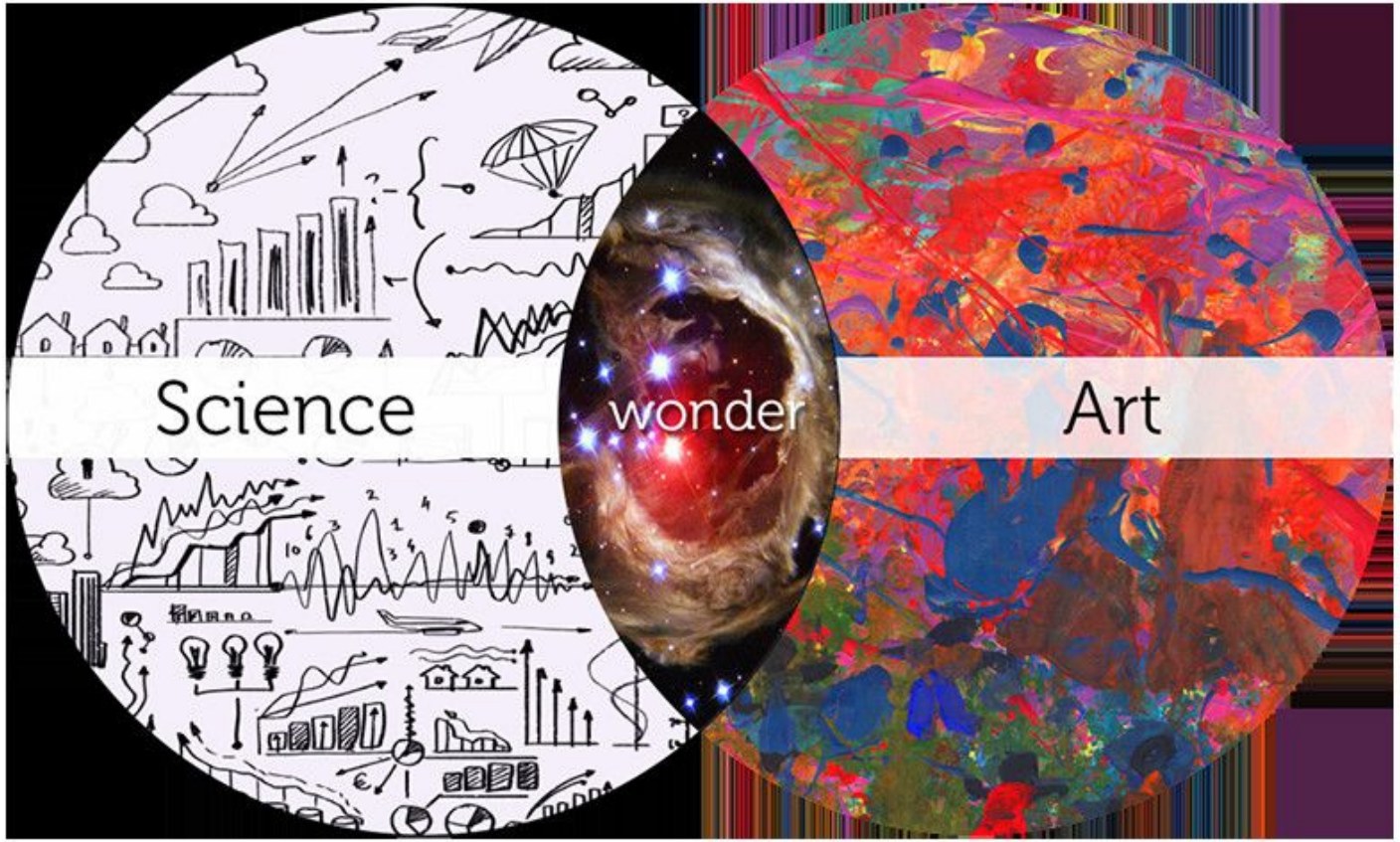


Science impacts the way we view ourselves





Science and the Arts influence each other



Science impacts our economy and our politics



This course is also for science students



What are we going to do in this course?

Each class from **6pm to 7:30pm Wednesday** will comprise:

1. A short informational lecture (20 minutes)
2. In class activities and discussion (20 minutes)
3. Q&A with a guest speaker (45 minutes)

We have just seven sessions

Wednesday September 9, 2026

- Week 1 Lecture: What is science? The history and philosophy of science.
- Week 1 Guests: **Michael Strevens** (NYU Professor, Philosophy)

Wednesday September 16, 2026

- Week 2 Lecture: Who should pay for science? The economics of science funding.
- Week 2 Guests: **Nate Williams** (The Sloan Foundation)

Wednesday September 23, 2026

- Week 3 Lecture: How science is communicated. From science publishing to the popular press.
- Week 3 Guests: **Apoorva Mandavilli** (Science Journalist, The New York Times)

Wednesday September 30, 2026

- Week 4 Lecture: The dark side of science. Eugenics, the atomic bomb, and fomenting doubt.
- Week 4 Guests: **Nina Gray** (NYU Vice Provost of Research)

Wednesday October 7, 2026

- Week 5 Lecture: When should we believe science? The replication crisis, fraud, and retractions.
- Week 5 Guests: **Ivan Oransky** (NYU Distinguished Journalist in Residence, Arthur Carter Journalism Institute & co-founder of Retraction Watch)

Wednesday October 14, 2026

- Week 6 Lecture: Politics and science: mistrust, misinformation, and polarization.
- Week 6 Guests: **Pat Egan** (NYU Politics & Director of NYU Berkley Institute for Civil Discourse & Civic Solutions)

Wednesday October 21, 2026

- Week 7 Lecture: The future of science. How can we use science to solve large societal problems?
- Week 7 Guests: **Michael Purugganan** (NYU Dean of the Faculty of Arts and Science & Professor of Biology)

Expectations

Assignments (60%)

A crucial component of this class is your ongoing engagement with course materials and with each other through homework assignments. These assignments will take different forms: posts on a discussion forum, video annotations using Annoto, quizzes, etc. The goal of these assignments is for you to deepen your engagement with course materials and to serve as a tool for dialogue and interaction with your classmates and me.

Final Project (20%)

For your final project, you'll explore a specific question or topic that interests you and apply the key concepts we've discussed throughout this course. You can complete this project individually or in small groups of two to three students.

You can present your findings in a variety of creative formats: a recorded narrated presentation, a video, a podcast. You will also submit a 1-page individual reflection. Regardless of the format, your project should demonstrate a thoughtful analysis that is grounded in the course material. It should also be fun!

1 credit, pass/fail

To receive a **Pass** for the course, you must meet all of the following criteria:

- Attend at least 6 out of the 7 scheduled sessions.
- Actively participate in class discussions and activities.
- Complete and submit all homework assignments and the final project by the stated deadlines.
- The final project must earn a Satisfactory or Excellent rating based on the project rubric. A project that receives a Needs Improvement rating will not be considered for a Pass until it has been revised and resubmitted to a passing level. You may resubmit once.

Both IRL and tech forward

Use of digital devices in the classroom

Access to the Internet is a valuable aid, but these technologies can be a distraction. We will use a 'Lids Down' policy during lectures to ensure deep thinking. When we move to poll questions to gauge understanding, data analysis or research activities, I will invite you to bring your devices out. At all other times, we prioritize eye contact and active listening.

Let’s learn a bit about who you are…

What year are you in?

(A) Freshman

0%

(B) Sophomore

0%

(C) Junior

0%

(D) Senior

0%

(E) Grad

0%

(F) Other

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What school are you in?

CAS 0%

Tandon 0%

Stern 0%

Gallatin 0%

Steinhardt 0%

Tisch 0%

Global Liberal Studies 0%

Nursing 0%

Grossman 0

Courant 0

SEE MORE

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What are you planning to major in?

(A) Humanities

0%

(B) Social Science

0%

(C) Science

0%

(D) Math

0%

(E) Engineering

0%

(F) The Arts

0%

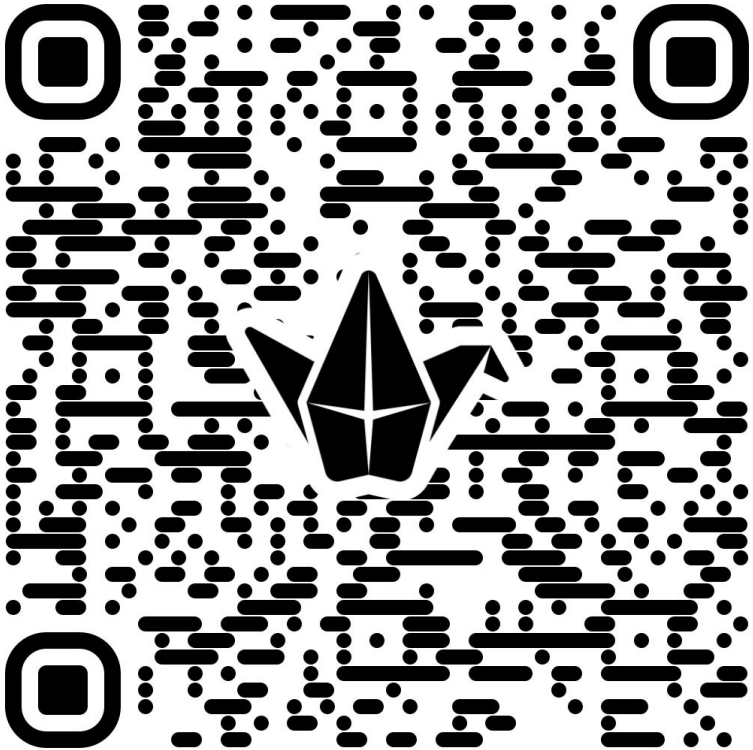


Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

This week’s assignment: introduce yourself on padlet

- What is your name?
- What's your major?
- What’s the best thing you read, watched, or listened to this summer?
- What is one thing that would make this class worth your time?


Padlet
dg107_2 • 5m
What's wrong with Science: Class introductions
Perth WA, Australia
Perth WA, Australia
Slideshow
Share
Post
North Pacific Ocean
North Atlantic Ocean
South Pacific Ocean
South Atlantic Ocean
Indian Ocean
Southern Ocean
ANTARCTICA
Greenland
Iceland
Canada
United States
Mexico
Russia
China
Japan
South Korea
India
Indonesia
Australia
New Zealand
Papua New Guinea
Brazil
Argentina
South Africa



https://padlet.com/dg107_2/what-s-wrong-with-science-class-introductions-s0236khoncojf1rw0z3f

Bonus assignment: Contribute to the class spotify playlist

